Regulation & Governance

Regulation & Governance (2017) ••, ••–•• doi:10.1111/rego.12143

Role perceptions and attitudes toward discretion at a decentralized regulatory frontline: The case of organic inspectors

David P. Carter*

Department of Political Science, University of Utah, Salt Lake City, UT, USA

Abstract

The complexity of decentralized regulatory contexts can threaten program fidelity, particularly if it results in divergence between program goals and the intents of regulatory inspectors. This paper investigates how inspectors negotiate the conflicting demands of a decentralized program by examining how they perceive their regulatory roles – the primary responsibilities that inspectors ascribe to their functions and the entities to which they feel responsible – and how these role orientations are related to inspectors' attitudes toward the use of discretion. The study findings indicate that in the decentralized administration of United States organic food regulations, inspectors experience multiple, and sometimes conflicting, role orientations. The presence of multiple role orientations, however, does not seem to affect how inspectors approach their responsibilities. The combined strengths of quantitative and qualitative data are leveraged to offer explanations for the study findings and identify avenues for future research.

Introduction

As the operative frontline of regulatory activity, the work of inspectors is an essential element of any regulatory program. Regulatory administration is a complex social phenomenon (Nielsen 2006), requiring inspectors to "be alternately informative, cajoling, educating, or punitive as needed to produce the desired levels of cooperation and compliance" (May & Wood 2003, pg. 118). The discretion that inspectors exercise can support regulatory effectiveness – if through "responsive regulation" inspectors take into account contextual and motivational considerations that differ across regulatees (Tyler 1990; Ayres & Braithwaite 1992). At the same time, too much flexibility in inspector activity can pose a threat to regulatory program fidelity if responsive actions undermine compliance goals or are perceived to reflect regulatory capture (May 2003; Pautz 2009; Pautz & Rinfret 2013).

Understanding the manners in which inspectors exercise discretion requires considering the various influences that they face, including signals embedded in the content of policy (May & Winter 2007), interactions with regulatees (Fineman 1998; Nielsen 2006, 2007), and inspectors' own value orientations (Hedge *et al.* 1988). Contemporary regulatory program designs impose additional and potentially conflicting pressures with which inspectors must contend. Movements to replace hierarchical program structures with administrative decentralization reliant on "regulatory intermediaries" (Levi-Faur & Starobin 2014) can both multiply and dilute the objectives that are communicated to frontline workers (Sandfort 2000; Buffat 2015). The increasing prevalence of nongovernmental organizations as regulatory administrators introduces organizational incentives that often diverge from program aims (Dias & Maynard-Moody 2007; Buffat 2015). Programs that treat program targets as public service customers can induce frontline workers to emphasize client desires over policy goals (Fountain 2001; Dubois 2010).

The complexity of decentralized policy contexts can pose a threat to program fidelity, particularly if the result is a divergence between program goals and the operative intents of frontline workers (Meyers *et al.* 2001). This paper examines how inspectors negotiate and internalize the potentially conflicting demands of a decentralized program context, and whether the responsibilities that inspectors prioritize shape their propensity to adhere more strictly or loosely to mandated regulatory directives. This is accomplished through two research objectives. The first objective is to understand how inspectors perceive their regulatory roles – the primary responsibilities that inspectors

Correspondence: David P. Carter Department of Political Science University of Utah Building 73, Room 223, 332 South 1400 East, Salt Lake City, UT 84112. E-mail: david.carter@mpa.utah.edu

Accepted for publication 29 October 2016.

© 2017 John Wiley & Sons Australia, Ltd

ascribe to their functions and the individuals or entities to which they feel responsible (Schuler, 1975; Fineman 1998; Selden *et al.* 1999). The second objective is to understand how these role orientations are related to inspectors' discretionary attitudes, understood as the extent to which inspectors believe regulatory action should closely adhere to regulatory program rules (Hill 2013).

Inspector role perceptions and discretionary attitudes are studied in the context of organic food regulation under the United States Department of Agriculture's (USDA's) National Organic Program (NOP). The NOP context is chosen for two reasons. First, the NOP is quasi-voluntary, as regulatees are only subject to regulatory oversight if they wish to market their products as "organic." Regulatees pay for program participation, introducing a tension between program participants as regulatees versus regulatory service clients (Conglianese & Lazer 2003). Second, although the NOP is charged with establishing national standards by which organic food production is regulated in the US, it relies on independent governmental and nongovernmental certification organizations to ensure certified organic operations' adherence to NOP regulations. The decentralized nature of NOP delivery raises the potential for the incentives and objectives that are introduced by independent administrative organizations to influence the manners in which inspectors approach their regulatory responsibilities (Sandfort 2000; Dias & Maynard-Moody 2007).

The study follows a mixed method research design, leveraging the combined strengths of quantitative and qualitative data to depict how organic inspectors understand their function (Creswell 1994). A survey quantifies inspector role perceptions and attitudes toward discretion. Interviews provide more nuanced descriptions of how inspectors understand their responsibilities, balance conflicting incentives and constraints, and exercise judgment while performing their regulatory duties. The study findings provide only limited support for the general expectation that inspector role perceptions shape, and are thereby likely associated with, inspectors' attitudes toward discretion. Drawing on implementation theory and contextual, social, and institutional characteristics of the NOP regulatory setting, the paper offers explanations for the findings and opportunities for future research.

Inspector role perceptions and attitudes toward discretion

Regulatory administration scholars have studied the work of inspectors from several angles. For example, May and Winter (1999, 2000; Winter & May 2001; see also May & Wood 2003, Pautz & Rinfret 2013) focus on conceptualizing and measuring inspectors' monitoring and enforcement activity, finding that inspectors' approaches can be usefully described by the degree of formalism and coercion that they exhibit. Other studies identify discrete influences on inspector behavior, such as policy designs (May 1993), political signaling (May & Winter 2007), and the values held and expressed by inspectors (Hedge *et al.* 1988). Similarly, Nielsen (2006, 2007) draws attention to the nature of inspector-regulatee interactions to explain inspectors' behavior, and the influence of factors such as regulatee reputation and power in shaping inspector practices.

This study takes a slightly different approach by focusing on the role orientations that inspectors express and the relationship that inspector role perceptions have with their attitudes toward discretion. Following role theory, inspector roles are defined as reflecting the primary responsibilities that inspectors ascribe to their functions and the individuals or entities to which they feel responsible (Biddle 1986; Selden *et al.* 1999). Roles, thus, reflect what Evans (2010, 2011) refers to as street-level bureaucrat "alliances" within an implementation structure. Drawing attention to how inspectors understand their roles moves beyond cataloging discrete influences on inspector behavior to document how inspectors internalize different influences within their professional setting (e.g. Hedge *et al.* 1988; Brehm & Gates 1999; May & Winter 2007). In doing so, it highlights the roles with which inspectors identify as possible mechanisms by which regulatory organizations, program contexts, and individual inspector values are translated into inspectors' discretionary attitudes.

Importantly, this study seeks to understand inspector role perceptions in a decentralized regulatory program, as such contexts introduce structural complexity that may further complicate inspector roles when compared with more traditional program contexts (Brodkin 2012). In traditional regulatory settings, inspector roles are assumed to be shaped by two dominant principals: the regulatory bureaucracies inspectors are accountable to and the individuals they are charged with regulating (e.g. Pautz, 2010). While inspectors are (by definition) regulatory actors, the extent to which they identify with regulatees can shift inspector orientations to align with regulatee concerns (Nielsen 2006, 2007; Evans 2010; Pautz & Rinfret 2013). Decentralized regulatory programs introduce another commitment by which inspectors' roles might be defined: in a context in which inspectors are responsible to independent organizations

 © 2017 John Wiley & Sons Australia, Ltd

charged with regulatory administration, inspectors are incentivized to internalize employer interests in addition to regulatory objectives (Hupe *et al.* 2015). This study therefore incorporates three possible role perceptions: inspectors as regulatory monitors, inspectors as service providers to regulatee operations, and inspectors as representatives of independent employer organizations.

Rather than develop a single uniform understanding of their regulatory function, inspectors are likely to perceive these multiple role orientations simultaneously, resulting in a "web of obligations, relationships, and meanings" that inspectors regularly navigate (Musheno & Maynard-Moody 2015, p. 169). Sometimes inspector role responsibilities might converge, for example, when regulatory program goals and the objectives of independent administrative organizations align. At other times, inspectors might perceive themselves to be caught between divergent obligations, such as when regulatory program directives and regulatee concerns compete for favorable consequences stemming from inspector decisionmaking. In these instances, it is likely that an inspector will resolve conflicting role orientations by prioritizing certain obligations over others (Evans & Harris 2004). Regardless of the extent to which inspector role perceptions most often align or conflict with one another, or which obligations inspectors prioritize, a baseline expectation is that inspectors operating in a decentralized regulatory program will be faced with navigating potentially diverging role perceptions:

Hypothesis 1: In a regulatory context characterized by administrative decentralization, inspectors will perceive multiple, sometimes conflicting, role orientations.

Beyond documenting the role orientations that inspectors perceive in a decentralized regulatory setting, this paper examines the relationship between inspectors' role perceptions and their discretionary attitudes. In this study, discretionary attitudes are assumed to reflect inspectors' comfort or aversion to the application of individual judgment in the execution of program requirements. In other words, the extent to which inspectors perceive what Hupe *et al.* (2015, p. 17) call "discretion-as-used" to be an appropriate element of inspectors' responsibilities. Role theory indicates that individual attitudes and behavior can be anticipated by the roles that individuals internalize (Biddle 1986). This paper tests the general expectation that inspectors' discretionary attitudes are shaped by, and will therefore be associated with, the regulatory role orientations with which inspectors more closely align. More specific hypotheses are facilitated by considering the objectives and interests of the entities associated with each of the inspector roles incorporated in this study: inspectors as regulatory monitors, employee representatives, and service providers.

First, the more inspectors prioritize their responsibilities to a regulatory program over their obligation to their employing organizations, or the regulatees that they interact with, the more likely they are to adhere to the "discretionary space" formally constructed by the regulatory framework in question (Buffat 2015). In emphasizing their regulatory obligations, such inspectors are likely to exhibit more "rule-bound" (May & Winter 2000), less accommodating (Pautz & Rinfret 2013) inspection practices. Whether it be the result of internalization of policy goals (Meyers *et al.* 2001), or a sense of public service duty, this close adherence to mandated regulatory directives is expected to be reflected in a relative aversion to the use of discretion:

Hypothesis 2: The more closely an inspector aligns with a regulatory monitor role orientation, the less likely he or she is to exhibit a favorable attitude toward the use of discretion.

Second, the more an inspector identifies as a representative of an independent administrative organization, the more likely they are to attend to organizational goals over strict adherence to policy directives. The multiplicity of organizational objectives that contend with program requirements likely favor a more flexible application of regulatory directives (Meyers *et al.* 2001) – unless organizational incentives are aligned with a strict adherence to program rules, for example, through performance parameters (Considine 2000; Brown *et al.* 2006). Perhaps most notably, in a program context where regulatees pay for program services, greater inspector discretion translates into a more accommodative regulatory approach through which administrative organizations might attract and retain regulatee clients (Conglianese & Lazer 2003). Inspectors that prioritize their obligations to their employer organizations, thereby aligning with an employee or representative role orientation, are expected to favor the use of discretion:

© 2017 John Wiley & Sons Australia, Ltd

Hypothesis 3: The more closely an inspector aligns with an employee representative role orientation, the more likely he or she is to exhibit a favorable attitude toward the use of discretion.

Finally, regulatory inspections are inherently social interactions, and the nature of the inspector-regulatee relationship can affect how inspectors carry out their responsibilities (May & Winter 1999; Nielsen 2006, 2007). Inspectors may favor cooperative interactions and even "partnerships" with regulatees over more adversarial relationships, particularly when trust and interdependence between inspectors and regulatees are high (Pautz & Rinfret 2013). Furthermore, the decisions that inspectors make can have significant consequences for regulatees (Nielsen 2015). In the case of organic inspections, for example, a farmer's access to the organic foods market, and the price premium that comes with it, depend on the results of annual inspection reports and the inspector reporting the decisions that comprise them. Inspectors that perceive themselves to be service providers to regulatory clients are assumed to be more likely to sympathize with regulatee interests and concerns, and are therefore more likely to be comfortable exercising judgment, and perhaps flexibility, in the application of regulatory standards:

Hypothesis 4: The more closely an inspector aligns with a service provider role orientation, the more likely he or she is to exhibit a favorable attitude toward the use of discretion.

Through these hypotheses, this study examines the relationship between inspector role perceptions and attitudes toward discretion in the context of organic food regulation in the US. The extent to which these theoretical expectations are likely to generalize beyond this context is touched on in the conclusion. The context is described in the next section, followed by a presentation of the study's mixed method design.

Study context and research design

This paper examines the role perceptions and attitudes toward discretion of inspectors charged with administering organic food regulations under the USDA's NOP. As required by the 1990 Organic Foods Production Act, operations (such as farms) that market and sell their products as "organic" must hold organic certification under the NOP and adhere to the national organic standards outlined in NOP regulations.[1] As such, the NOP can be considered a "quasi-voluntary" program – operations actively seek and participate in the NOP. Operations can avoid participation requirements by refraining from using the term "organic" in their product marketing and labeling.

The NOP's decentralized structure is modeled loosely after International Organization for Standardization (ISO) guidelines for accreditation and certification systems. The NOP codifies uniform national organic standards through administrative rulemaking, and accredits independent governmental, nonprofit, and for-profit organizations (certifiers) to certify operations according to the regulations. As of 2013, the NOP had accredited 48 US based certifiers: 17 private for-profit organizations, 11 nonprofits, and 20 from the public sector.[2] Public sector certifiers can be further broken down into three different forms: 17 fall under state government programs, such as the Colorado Department of Agriculture Organic Program; two fall under California county governments; and one operates out of Clemson University's Department of Plant Industry.

To attain organic certification, operations first submit an organic systems plan to the accredited certifier of their choosing, and pay them a fee for certification services. Once an organic systems plan has been reviewed and approved, the operation is inspected for congruence with the plan. These inspections are carried out on-site by an organic inspector (which is the focus of this paper). NOP regulations (2016) specify that the person who makes a final certification decision (i.e. whether certification is granted or not) cannot be the same individual who inspects an applicant operation. As a result, accredited certifiers either employ in-house inspectors, or contract with independent inspector personnel. Inspectors examine operation facilities, processes, and paperwork for congruence with the submitted organic systems plan and accordance with NOP regulations. Discrepancies with the plan or potential regulatory violations are then reported to the accredited certifier. According to NOP regulations, certification must be renewed once a year, which entails submission of changes to the organic systems plan and an annual on-site inspection.

The regulations require that certifier personnel exhibit sufficient expertise in organic production and handling to carry out certification duties, but stop short of specifying what types of education, experience, or training constitute "sufficient expertise" (NOP, 2016). As such, no formal requirements are specified for the education of organic inspectors. In practice, organic inspectors may attain training in two ways. First, inspectors that are hired as part-time

 © 2017 John Wiley & Sons Australia, Ltd

or full-time personnel from an accredited certifier may receive in-house training (Kahn & Scoles 2013). Second, many inspectors receive training from the International Organic Inspectors Association (IOIA), a professional association with the express objective of providing quality organic inspector education (International Organic Inspectors Association [IOIA] n.d.). According to Margaret Scoles, IOIA Executive Director: "[the IOIA is] the industry standard. There is no rule that requires that anyone take our training, but almost all inspector trainees have taken a basic course with us" (Kahn & Scoles 2013).

Reflecting the decentralized nature of NOP administration, no consolidated database exists on organic inspectors, and as a result, population characteristics are unknown. The IOIA lists 159 US inspector members; however, the IOIA is a voluntary professional organization representing an unspecified segment of the inspector population (IOIA n.d.). The composition of the study sample, and the manners in which inspectors were identified for the study, are provided in the remainder of this section.

Research design

This paper examines how organic inspectors under the NOP understand and approach their regulatory responsibilities – subjective experiences that are best understood by asking the inspectors themselves (Creswell 1994). To construct the study sample, organic inspectors were identified through a combination of targeted sampling based on organizational affiliations and snowball sampling techniques (Atkinson & Flint 2001). Inspectors were first identified through the list of IOIA inspector members and Internet-based searches for inspectors identified on the websites of accredited certifiers. As the study progressed, and in particular during the interviews, participants verified as NOP regulation inspectors were asked to provide the contact information of colleagues. Certifier administrators were also contacted, and requests were made for the contact information of current employee and contract inspector personnel. In total, the names and contact information of 260 individuals thought to be NOP regulation inspectors were collected; 159 were identified as IOIA members.

The study followed a parallel mixed method research design in which qualitative and quantitative data were collected simultaneously (Leech & Onwuegbuzie 2007). The design merges inductive and deductive approaches (Van de Ven 2007). In doing so, it combines the strengths of qualitative data, such as nuanced research findings and the identification of important considerations not included in a deductive framework (Gerring 2012), with the strengths of quantitative data: greater transparency and replicability, and data collection from a larger sample of participants (Creswell 1994; Teddlie & Tashakkori 2009). Quantitative data were collected through an online survey of potential organic inspectors. Qualitative data were collected through semi-structured interviews conducted with an inspector subsample.

Organic inspector survey. An email request for participation in the online survey was sent to the 260 identified potential organic inspectors. In total, 107 responses were received, 10 of which indicated that the respondents no longer served as organic inspectors. Another five surveys were missing sufficient data to render the results unusable, resulting in 92 serviceable surveys (35 percent of the initial 260 requests sent). The survey first asked questions regarding respondents' employment status, such as whether they were part-time, full-time, or contract employees. Respondents were also queried as to their organizational affiliations: the number of certifiers they worked for, and whether they worked for government or private (nonprofit or for-profit) certifiers. The breakdown of respondents by work status and organizational affiliation is provided in Appendix I, alongside additional sample characteristics.

The primary survey data were gathered in two question batteries. As the first objective of this study was to understand how inspectors perceive their roles, a battery asked for respondents' level of agreement with three role identifications: (i) regulatory monitor for the NOP, (ii) employee and representative of the certifier(s) the inspector worked for, and (iii) a service provider to the operation under inspection. The second study objective was to ascertain inspectors' attitudes toward discretion. A battery asked for respondents' agreement with statements pertaining to three aspects of inspector discretion: (i) discretion in the interpretation of NOP regulations, (ii) discretion in reporting regulatee bookkeeping issues, and (iii) discretion in reporting potential noncompliance issues.

Organic inspector interviews. Semi-structured interviews were conducted via telephone with a subsample of 10 organic inspectors. The subsample frame was purposeful and designed with the goal of achieving adequate representation of the

© 2017 John Wiley & Sons Australia, Ltd

most evident population sub-groups (referred to as a parallel sampling frame by Onwuegbuzie and Leech 2007), including whether the inspector was employed on a part/full-time basis or as a contractor, and whether the inspector worked for governmental or private (nonprofit or for-profit) certifiers. Because of a higher interview request response rate among government-affiliated inspectors, five interview participants were part-time or full-time employees of government certifiers, two were contract inspectors that worked for both government and private certifiers, and three were contract inspectors to exclusively private certifiers.

With the exception of overrepresentation among governmental employee inspectors, the interview respondents reflected survey sample characteristics fairly well (see Appendix I). Five of the 10 interviewees had been conducting organic inspections for over 10 years. One was a new organic inspector as of the preceding year, and the four remaining inspectors had conducted organic inspections for between four and six years. Four interviewees were women, and six were men. The interviewees were not asked for their exact age, however, two were over 65 years old, six were identified as "middle aged," and the remaining two inspectors were under 30 years old.

Interview questions focused on respondents' experiences and perceptions of the inspector function in organic food regulation. A question specifically asked how respondents saw their role within the NOP context.[3] The interviews were recorded, transcribed, and uploaded to QDA Miner software to facilitate qualitative analysis. A modified constant comparison approach was applied to the interview data – transcripts were first coded for discussion related to inspector role perceptions and the application of discretion (Glaser & Strauss 1967; Leech & Onwuegbuzie 2007). Coded data were then iteratively reviewed for themes within these general topics that emerged across the interviews (Creswell 1994; Leech & Onwuegbuzie 2007).

The approach appropriately provided descriptive detail to quantitative survey findings and identified narrative explanations for study findings, as presented in the Discussion section of this paper (Gerring 2012). As a result of the overrepresentation of governmental employee inspectors in the interviewee sample, interviewees' organizational affiliations and employment status were taken into consideration when qualitative data were considered for inclusion in this paper – in particular, in the case of direct quotations. Unique interviewee identifiers were assigned to each interview transcript, and reported in the paper to ensure that a minority of interviewee comments did not disproportionately influence the findings.[4]

Results

The first research objective of this paper was to describe how inspectors perceive their regulatory roles, with the baseline expectation that in a decentralized regulatory context, such as the NOP, inspectors will perceive multiple, sometimes conflicting, role orientations. This expectation is examined in the first part of this section. The second research objective – to examine the relationship between inspector role perceptions and attitudes toward discretion – follows.

Organic inspector role perceptions

Survey respondents' role perceptions were studied through a question battery that asked the extent to which respondents identified with three roles: regulatory monitor for the NOP, an employee and representative for the certifier(s) they work for, and a service provider to the operation under inspection. As displayed in Table 1, on average, respondents most strongly identified as regulatory monitors for the NOP (average = 3.9/5), followed by representatives of certification employers (3.5/5). Survey respondents were least likely to identify as service providers to regulated operations (2.5/5).[5] Collectively, the results suggest that the average organic inspector perceives themselves as

Table 1 Inspector role perception survey findings

Role	Description	Mean response (SD)
Regulatory monitor	Regulatory monitor for the NOP	3.9(1.1)
Employee/ representative	Employee and representative of the certifier(s) the inspector works for	3.5(1.2)
Service provider	Service provider to the operation under inspection	2.5(1.2)

Mean number represents respondents' agreement with role orientation: 1 = "strongly disagree;" 2 = "disagree;" 3 = "neither agree nor disagree;" 4 = "agree;" 5 = "strongly agree." NOP, National Organic Program; SD, standard deviation.

 © 2017 John Wiley & Sons Australia, Ltd

performing two of the three roles queried – regulatory monitor and organizational representative – supporting the expectation that inspectors in a decentralized regulatory context will experience multiple role orientations.

The interview findings corroborated the survey results. While all three role perceptions were expressed during the interviews, as exhibited in the quotes presented in Table 2, regulatory monitor was the most frequently cited role identification, referenced in all 10 interviews. Employee and representative of certifiers was the second most frequently cited role (cited in seven interviews), and service provider was the least frequently cited (cited in five interviews). The comments of one interviewee seemed to summarize the average inspector's role orientation succinctly, as "a kind of nested egg thing:" the inspector first saw her role as a regulatory monitor.[6] Within the regulatory monitor role, she was secondarily a certifier's employee, and within that role she provided a service to organic operations.

Beyond the expectation that inspectors would perceive multiple role orientations, it was hypothesized that inspectors would at times experience conflict between their different roles. Interview respondents readily recognized role tensions that at times required them to negotiate competing expectations. In all of the examples provided by interviewees in which differing expectations were encountered, the responsibility that inspectors ascribed to their role as regulatory monitors appeared to supersede conflicting pressures. For example, one contract inspector interviewee described encountering a situation in which the inspector's judgment of a noncompliance issue differed from that of their employing certifier's:

> This is what the certifier said: 'this is a dilemma between service and regulatory.' I had a notion to tell him, well I think your first job is regulatory service, I mean be polite and all that stuff and give them timeliness, but you are not going to give in and call it customer service. If they are not doing something right, you have to be stern.[7]

Importantly, an additional aspect of inspectors' regulatory monitor role emerged from several interviews. A number of interview participants underscored the consumer protection aspect of the inspector function. As one government-affiliated interviewee stated: "We are also working to protect the…consumer. If they buy organic, it is organic, and is certified organic under the same set of standards, whether [the operation] is Starbucks or a half acre lavender grower who is a hobby farmer."[8] Consumer confidence in the organic label is an essential element of the organic industry, according to these interviewees, and the work of inspectors is fundamental to consumers' trust in "the system." For example, a contract inspector interviewee reflected on his role: "[inspectors] are really the only ones…out there, that have direct contact with most producers…I represent the consumer, ultimately."[9]

The relationship between organic inspector role perceptions and discretionary attitudes

The second objective of this paper was to examine how inspectors' role orientations are related to their attitudes toward the use of discretion. Survey respondents' attitudes toward discretion were measured through queries pertaining to inspector interpretation of the NOP regulations, the reporting of issues with regulatee recordkeeping, and the reporting

Table 2 Interview quotes representing coded inspector role perceptions

Role	Quote
Regulatory monitor	"First and foremost, I am there to regulate and assess – are you complying with a national set of standards?" (Interview ID10)
Regulatory monitor	"…we are providing a service to the producer, obviously, but our responsibility, if you like, is to uphold the NOP. That's what we are doing." (Interview ID1)
Employee/representative	"I am definitely also an employee of the [certifier] as an inspector, so unless I was asked to do something clearly unethical I am responsible to them." (Interview ID2)
Employee/representative	"I provide a service to the certifier so that they can in turn – can perform their regulatory role." (Interview ID9)
Service Provider	"I think we are providing a service to organic farmers… so they can keep their organic certifications." (Interview ID3)
Service Provider	"I see it as a service to the people we inspect because we are a state organization with education involved and that we really do focus – we go to a lot of meetings and outreach to make sure people understand what is required to be certified." (Interview ID9)

© 2017 John Wiley & Sons Australia, Ltd

of potential noncompliance. As indicated in Table 3, the average respondent agreed that organic inspectors should follow the NOP regulations to the letter and that every potential instance of noncompliance should be reported. Similarly, the average respondent disagreed with the idea that recordkeeping inconsistencies could be overlooked if organic integrity is not threatened. Collectively, the results suggest relative opposition among survey respondents to the use of discretion in the identified manners. These results were corroborated by interview findings, represented through comments such as: "I go through the points and mark everything on the report, if there were any issues of concern"[10] and "I write up everything."[11]

Correlation analysis was performed to examine the relationship between inspector role perception and discretionary attitude survey measures. The results are displayed in Table 4. The first hypothesis regarding this relationship predicted that the more an inspector aligns with a regulatory monitor role, the less likely he or she would be to hold favorable attitudes toward the use of discretion. Supporting the expectation, when respondents identified as NOP regulatory monitors, they took an unfavorable attitude toward the application of discretion. The support is limited, however, by correlation coefficients that suggest relatively weak associations ($r = -0.09$ to -0.21), and only one statistically significant finding. The second hypothesis anticipated that greater alignment with an organizational employee/representative role would be associated with a more favorable attitude toward the use of discretion. No support was found for the expectation, as the findings indicate virtually no relationship between inspectors' alignment with the employee/representative role and their discretionary attitudes. In contrast, respondent identification with a service provider role was positively associated with favorable discretionary attitudes across all three measures, supporting the final hypothesis. Correlation coefficients indicated weak-to-moderate relationships between the service provider role and discretionary attitudes ($r = 0.18$ to 0.38) that were statistically significant at $P < 0.10$ to 0.01.

Interview data shed further light on organic food inspectors' attitudes toward discretion, and how these are related to inspectors' understanding of their regulatory function. First, interview comments suggested that respondent opposition to discretion is tied to the fact that NOP regulations mandate a clear distinction between inspectors as monitors and certifier organizations as regulatory administrators. Several interviewees emphasized that it is the duty of inspectors to report on regulatees' practices, and that judging the importance of potential noncompliance, or even determining whether an issue represents noncompliance, is the responsibility of certifiers. This sentiment was articulated well in the pronouncement: "As inspectors we are instructed to report findings rather than make decisions or express opinions."[12] Another interviewee stated: "As an inspector it is important to understand what our role is, and what our scope is – I don't make certification decisions."[13]

Second, a NOP regulation conflict of interest provision stipulates certifiers (and by extension, inspectors) refrain from activities that might be considered "consultation" while administering regulations. Interviewees reported that as a result of this provision, they limited the flexibility that they exhibited when interacting with regulatees, at times against their own inclinations or in manners that they felt were unproductive. For example, an employee inspector of a government certifier compared inspecting for NOP regulations with administering other regulatory standards, stating: "I always feel rather constrained, you know…in our other roles, we are allowed to help people, to problem solve, but we can't do that as inspectors for the NOP."[14] Another interviewee lamented the "misfortune" that the conflict of interest provision represented for smaller organic operations. According to this interviewee, inspectors are often "the

Table 3 Inspector attitudes toward discretion survey findings

Discretion type	Measure	Mean response (SD)
Interpretive discretion†	Organic inspectors should follow the NOP regulations to the letter.	4.0 (0.8)
Reporting discretion – recordkeeping	The organic inspector should not report an operation's inconsistent documentation if organic integrity is not threatened.	1.9 (0.9)
Reporting discretion – noncompliance†	Every potential noncompliance issue should be reported by organic inspectors without exception.	3.9 (1.0)

†Survey response scale reversed to represent a measure of favorable attitude toward discretion; table reports non-transmuted values. Mean value is respondents' agreement with the statement: 5 = "strongly agree" and 1 = "strongly disagree." NOP, National Organic Program; SD, standard deviation.

 © 2017 John Wiley & Sons Australia, Ltd

Table 4 Associations between role perception and attitude toward inspector discretion

	Interpretive discretion†	Reporting discretion – recordkeeping	Reporting discretion – noncompliance †
Regulatory monitor	-0.16	-0.21**	-0.09
Employee/representative	0.01	0.04	0.17
Service provider	0.18*	0.24**	0.38***

*$P < 0.10$; **$P < 0.05$; ***$P < 0.01$.
†Survey response scale reversed such that higher values reflect a favorable attitude toward discretion.

only people that [smaller operations] would ever see from the organic industry," and without direction and assistance from inspectors "they are completely out there on their own."[15]

Discussion

This paper examined how, in the context of a decentralized program context, inspectors understand their regulatory roles, and how these role perceptions are related to their discretionary attitudes. The findings suggest that a measured relationship exists between how inspectors perceive their roles and how they feel toward the use of inspector discretion: the more an inspector identifies as a regulatory monitor, the more likely he or she is to favor a strict adherence to program rules. In contrast, the more an inspector identifies as a regulatory service provider, the more likely he or she is to favor the use of discretion in a manner that might deviate from regulatory directives.

The first theoretical implication drawn from these findings is that the manners in which inspectors perceive their roles may be an overlooked factor in understanding inspection styles. Much of the existing literature assumes the nature of inspector-regulatee interactions is a strategic choice of inspectors seeking maximum compliance from regulated entities (e.g. May & Winter 2000; Pautz & Rinfret 2013). The quantitative and qualitative findings presented here indicate that inspectors' attitudes toward the use of discretion, and, by extension, the rigidity or accommodation with which they approach their inspection duties, may also be related to how they prioritize their obligations and the individuals or entities that they are principally responsible to (Evans & Harris 2004).

Despite the tension implied by discretionary attitudes that vary across different inspector role orientations, and qualitative data indicating that inspectors sometimes do experience such tensions, the results of this study suggest overall consistency in how American organic food inspectors understand and approach their functions. In a program context where regulatory administration is delegated to independent governmental and private sector organizations, and inspection duties are conducted by both in-house and contract employees, study findings depict role orientations that emphasize the regulatory aspect of inspectors' responsibilities and general opposition to the use of inspector discretion. The second theoretical implication stemming from these findings is thus support for a growing body of empirical research that counters "conventional" predictions of street-level divergence from policy goals (Sandfort 2000; Brodkin 2012). Following the suggestion of an anonymous reviewer – as evidence continues to accumulate that inspectors and other street-level bureaucrats may act as "principled agents" (see also Brehm & Gates 1999) more often-than-not, it may be time to begin seriously questioning why, and through what mechanisms, such policy commitment is realized.

To this end, qualitative study findings lend support to the theorized roles that several implementation context elements may contribute to consistency in regulatory program administration. First, aligning with social perspectives on the regulatory process (e.g. Fineman 1998; Nielsen 2006), Sandfort argues that front-line workers develop social schemas – "norms, collective beliefs, and shared knowledge" (2000, p. 742) – that transcend organizational and contextual differences, resulting in shared standards for street-level action. Consistency among organic inspector role perceptions could thus be the product of inspector norms established through formal mechanisms, such as the IOIA, and associated inspector interactions. As explained by Hill, nongovernmental implementation resources – such as professional associations – often carry the legitimacy and influence of peer authority figures, and thus "may constitute an unusually effective instrument for policy achievement" (2003, p. 280). This explanation was supported by the comments of several interviewees who singled out IOIA events as influential in how they approach their inspection

© 2017 John Wiley & Sons Australia, Ltd

practices, through what one interviewee described as opportunities "to interact with peers to gain knowledge of on-the-ground variations in organic practice and discuss handling of real-world challenges."[16]

A second factor that may help to explain the study findings is contingent on the extent to which organic inspectors assume regulatory monitor roles and oppose applications of inspector discretion because they condone and wish to support the NOP. Such an explanation maintains that consistency in frontline behavior may be attributable to inspector values and objectives that align with policy goals (Brehm & Gates 1999). This value-congruence explanation of study findings is supported by interview comments that highlight the importance of the consumer protection function of NOP regulations and identify inspectors as ultimately representing consumers.[17] This explanation rests not only on value congruence, but also an alignment of interests. If the organic industry, including inspectors' livelihoods, is dependent on public perceptions of regulatory integrity, then it is in the interest of inspectors to perform their duties in a manner that promotes consumer and citizen confidence in the NOP. This is likely to be especially true if inspectors feel that their behavior is open to scrutiny – if the sentiment expressed by one interviewee is shared among his peers: "We work in a fish bowl."[18]

A final set of factors, reflecting a top-down approach to understanding policy implementation (Mazmanian & Sabatier 1983), suggests that formal institutional constraints imposed by the design of NOP regulations may have the intended effects of aligning inspector orientations with a regulatory monitor function and limiting the application of inspector discretion. Evidence of two influential regulatory design elements came through in the study findings. The first is the distinction between inspectors as personnel responsible for monitoring certified operations and certifiers as the entities responsible for regulatory decisions, officially leaving the application of judgment "up to the certifier."[19] The second is the conflict of interest provision that prohibits inspectors from taking on an advisory role with regulated operations in the course of their regulatory duties. This provision may thus effectively inhibit inspectors from aligning too closely with regulatee concerns and interests to the extent that it undermines regulatory program fidelity (Conglianese & Lazer 2003).

Some caveats and study limitations

This study leveraged the highly decentralized case of US organic food regulation to investigate inspector role perceptions and discretionary attitudes. This case provided an appropriate context in which to examine how inspectors navigate the tensions stemming from conflicting obligations; however, it also calls for caveats regarding the generalizability of the study findings. For example, the multiple roles incorporated in this study – particularly inspector-as-employee and inspector-as-service provider – may not be applicable in more traditional regulatory structures, where regulatees do not opt into regulatory oversight and where inspectors report directly to a regulatory agency. In addition, and as suggested above, several additional facets of the organic regulation context likely contributed to the results. With this in mind, the findings would be most appropriately compared to similarly decentralized regulatory contexts. Furthermore, this study's results will prove most fruitful when the proposed explanations are tested in analogous contexts to better understand to what degree, and how, the identified factors contribute to program fidelity.

In addition to these caveats, the findings should be considered with an appreciation of three central limitations inherent in the study design and data. First, the fact that the population of inspectors that administer NOP regulations is undetermined makes it difficult to assess the extent to which study participants adequately represent the US organic inspector body. To mitigate this limitation, established strategies for sampling unidentified populations were used, including targeted and snowball sampling (Atkinson & Flint 2001). Nonetheless, the effects of inadvertently omitting potential participants on study findings are ultimately unknown. Second, the study relied on data from a sample of 92 survey respondents and 10 interview participants. As with any research that depends on voluntary participation, study findings may be subject to some degree of nonresponse bias (Cresswell 1994).

Third, and perhaps most importantly, the study relied on inspector self-reporting of role orientations and attitudes toward discretion. This approach, while arguably the most appropriate method for understanding inspectors' subjective experiences and perceptions (Creswell 1994), is particularly susceptible to social desirability bias in which study participants shape their responses to fit the attributes and behaviors that they believe to be most socially acceptable (Podsakoff *et al.* 2003). In addition, asking respondents to identify their role orientations and discretionary attitudes in the abstract does not account for contextual variances; for example, if inspectors' role orientations vary between regulatees perceived as "trustworthy" or "deceptive," or if discretionary attitudes vary between routine situations and

 © 2017 John Wiley & Sons Australia, Ltd

exceptional circumstances or regulatory grey areas. The study's mixed method approach, combined with guaranteed anonymity for study participants, were used to help limit the effects of social desirability bias (Leech & Onwuegbuzie 2007); however, alternative methods are needed to fully assess the extent to which social desirability influences self-reported study findings. Future research will also benefit from more nuanced measures of inspector perceptions and attitudes that capture how the phenomena vary based on contextual factors.

Despite these limitations, this paper presents a constructive step in understanding how individuals operating at the frontline of decentralized regulatory program structures perceive their responsibilities. The contributions of the paper extend beyond a mixed method case study of inspectors operating in a decentralized regulatory setting – qualitative data offered evidence of factors that help to explain the study's findings. These explanations serve as opportunities for future research into how inspectors understand and approach their regulatory functions.

Conclusion

This study was conducted to examine how inspectors operating in a decentralized program context understand and approach their regulatory responsibilities through two objectives. The first was to understand how inspectors perceive their roles. The second was to understand how these role perceptions are related to inspectors' attitudes toward the application of discretion. The effort was motivated by concerns that a structurally complex, decentralized program context may threaten regulatory program fidelity by diluting or otherwise undermining the program objectives that inspectors are responsible for implementing (Sandfort, 2000; Meyers *et al.* 2001).

Study results indicate little support for such concerns in the context of US organic food regulation. While the findings suggest that organic inspectors do perform multiple roles, and sometimes face competing expectations, these facts do not seem to impact how inspectors understand and approach their responsibilities in a manner that would weaken program objectives. In general, organic inspectors appear to see themselves first as regulatory monitors of national organic standards, second as representatives of their employing certification organizations, and third as service providers to regulatee operations. In addition, organic inspectors seem to oppose the application of inspector discretion when it comes to regulatory interpretation or the reporting of recordkeeping and noncompliance issues.

The primary contribution of these results is the suggestion that while inspectors might experience commitments to multiple (and sometime conflicting) regulatory role obligations, these differing orientations need not undermine inspectors' opinions toward adherence to program directives. In addition to confirming a general lesson of street-level bureaucracy research, that frontline workers tend to support the policy objectives they are charged with realizing (Hupe *et al.* 2015), this finding further speaks to the growing literature on regulatory intermediaries. Decentralized program structures such as the NOP rely heavily on such intermediaries: regulatory actors "with the capacity to affect, control, and monitor relations between rule-makers and rule-takers" (Levi-Faur & Starobin 2014, p. 21). This study contributes to this literature with the suggestion that, instead of altering regulatory directives through their own interpretations (see Talesh 2015), inspector intermediaries (and perhaps the legally mandated intermediaries they work for) can faithfully adhere to program rules, even when they may face incentives to do otherwise.

Combining existing theory with qualitative interview findings, this paper suggested a number of factors that likely influence how inspectors conceptualize their position within a decentralized regulatory system. First, formal inspector interaction venues, such as the IOIA, likely promote consistency in how inspectors understand their roles. Second, a context in which regulatory program integrity aligns with inspector values and interests likely promotes inspector identification as regulatory monitors over other potential role orientations. Third, in a context where inspector discretion is subject to multiple demands, two formal regulatory design elements may effectively limit the discretion exercised by inspectors: (i) making a formal distinction between the inspectors as reporting entities and an independent decisionmaking entity that applies judgment, and (ii) regulatory provisions that prohibit "consultation" or "advisory" behavior.

These explanations offer potentially fruitful avenues for a better understanding of how inspectors understand and approach their responsibilities in complex contemporary regulatory settings. The increasing prevalence of regulatory program characteristics such as those observed in US organic food regulation present additional contexts for research assessing these propositions, from governmental and private sector administration of building regulations (van der Heijden 2008), to decentralized administration of forestry certification standards (Meidinger 2006). An equal contribution is promised by an examination of the relationship that inspector role perceptions and attitudes toward discretion

© 2017 John Wiley & Sons Australia, Ltd

have with inspector practices and behavior. Such extensions of the research reported here offer a fuller understanding of the manners in which inspectors conceptualize and execute their regulatory responsibilities, and the manners in which inspector role orientations serve to sustain or undermine regulatory goals.

Acknowledgments

This research was funded by the National Science Foundation, grant #1124541. I would like to express my gratitude to Christopher M. Weible and John Brett (University of Colorado Denver), Saba Siddiki (Indiana University-Purdue University Indianapolis), Xavier Basurto and Ainsley Smith (Duke University), and Sara Miller-Chonaiew (Oregon State University) as collaborators in data collection efforts. I extend a particular thanks to Alison Kent (University of Colorado Denver) for much of the data collection that made this paper possible, and to Peter J. May, Kristin Olofsson, and Daniel Costie for helpful comments on early manuscript drafts.

References

Atkinson R, Flint J (2001) Accessing Hidden and Hard-to-Reach Populations: Snowball Research Strategies. *Social Research Update* 33, 1–4.

Ayres I, Braithwaite J (1992) *Responsive Regulation: Transcending the Deregulation Debate*. OUP, Oxford.

Biddle BJ (1986) Recent Developments in Role Theory. *Annual Review of Sociology* 12, 67–92.

Brehm J, Gates S (1999) *Working, Shirking, and Sabotage: Bureaucratic Response to a Democratic Public*. University of Michigan Press, Ann Arbor.

Brodkin EZ (2012) Reflections on Street-Level Bureaucracy: Past, Present, and Future. *Public Administration Review* 72, 940–949.

Brown TL, Potoski M, Van Slyke DM (2006) Managing Public Service Contracts: Aligning Values, Institutions, and Markets. *Public Administration Review* 66, 323–331.

Buffat A (2015) When and Why Discretion is Weak or Strong: The Case of Taxing Officers in a Public Employment Fund. In: Hupe P, Hill M, Buffat A (eds) *Understanding Street-Level Bureaucracy*, pp. 79–96. University of Chicago Press, Chicago.

Coglianese C, Lazer D (2003) Management-based Regulation: Prescribing Private Management to Achieve Public Goals. *Law & Society Review* 37, 691–730.

Considine C (2000) Selling the Unemployed: The Performance of Bureaucracies, Firms, and Non-profits in the New Australian "Market" for Unemployment Assistance. *Social Policy & Administration* 34, 274–295.

Creswell JW (1994) *Research Design: Qualitative & Quantitative Approaches*. SAGE, Thousand Oaks, CA.

Dias JJ, Maynard-Moody S (2007) For-profit Welfare: Contracts, Conflicts, and the Performance Paradox. *Journal of Public Administration Research and Theory* 17, 189–211.

Dubois V (2010) *The Bureaucrat and the Poor: Encounters in French Welfare Offices*. Ashgate, Farnham, UK.

Evans T (2010) *Professional Discretion in Welfare Services: Beyond Street-level Bureaucracy*. Ashgate, Farnham, UK.

Evans T (2011) Professionals, Managers, and Discretion: Critiquing Street-Level Bureaucracy. *The British Journal of Social Work* 41, 368–386.

Evans T, Harris J (2004) Street-level Bureaucracy, Social Work and the (Exaggerated) Death of Discretion. *British Journal of Social Work* 34, 871–895.

Fineman S (1998) Street-level Bureaucrats and the Social Construction of Environmental Control. *Organization Studies* 19, 953–974.

Fountain JE (2001) Paradoxes of Public Sector Customer Service. *Governance* 14, 55–73.

Gerring J (2012) *Social Science Methodology: A Unified Framework*. CUP, New York.

Glaser BG, Strauss A (1967) *The Discovery of Grounded Theory: Strategies for Qualitative Research*. Aldine, Chicago.

Hedge DM, Menzel DC, Williams GH (1988) Regulatory Attitudes and Behavior: The Case of Surface Mining Regulation. *Western Political Quarterly* 41, 323–340.

Hill HC (2003) Understanding Implementation: Street-level Bureaucrats' Resources for Reform. *Journal of Public Administration Research and Theory* 13, 265–282.

Hill M (2013) *The Public Policy Process*, 6th edn. Pearson, Harlow, UK.

Hupe P, Hill M, Buffat A (2015) *Understanding Street-level Bureaucracy*. University of Chicago Press, Chicago.

IOIA (International Organic Inspectors Association) (n.d.) International Organic Inspectors Association [Association website]. [Last accessed 19 Aug 2016.] Available from: http://www.ioia.net/

Kahn B, (interviewer), Scoles M, (interviewee) (2013) Margaret Scoles on Organic Inspection [Recorded interview]. [Last accessed 28 Nov 2016.] Available from: http://www.ioia.net/press_releases.html

 © 2017 John Wiley & Sons Australia, Ltd

Leech NL, Onwuegbuzie AJ (2007) An Array of Qualitative Data Analysis Tools: A Call for Data Analysis Triangulation. *School Psychology Quarterly* 22, 557–584.

Levi-Faur D, Starobin SM (2014) Transnational Politics and Policy: From Two-way to Three-way Interactions. Jerusalem Papers in Regulation & Governance, Working Paper 62. Hebrew University, Jerusalem.

May PJ (1993) Mandate Design and Implementation: Enhancing Implementation Effort and Regulatory Styles. *Journal of Policy Analysis and Management* 12, 634–663.

May PJ (2003) Performance-based Regulation and Regulatory Regimes: The Saga of Leaky Buildings. *Law & Policy* 25, 381–401.

May PJ, Winter S (1999) Regulatory Enforcement and Compliance: Examining Danish Agro-environmental Policy. *Journal of Policy Analysis and Management* 18, 625–651.

May PJ, Winter S (2000) Reconsidering Styles of Regulatory Enforcement: Patterns in Danish Agro-environmental Inspection. *Law & Policy* 22, 144–173.

May PJ, Winter SC (2007) Politicians, Managers, and Street-level Bureaucrats: Influences on Policy Implementation. *Journal of Public Administration Research and Theory* 19, 453–476.

May PJ, Wood RS (2003) At the Regulatory Front Lines: Inspectors' Enforcement Styles and Regulatory Compliance. *Journal of Public Administration Research and Theory* 13, 117–139.

Mazmanian DA, Sabatier PA (1983) *Implementation and Public Policy*. University Press of America, Lanham, MD.

Meidinger E (2006) The Administrative Law of Global Private-Public Regulation: The Case of Forestry. *European Journal of International Law* 17, 47–87.

Meyers MK, Riccucci NM, Lurie I (2001) Achieving Goal Congruence in Complex Environments: The Case of Welfare Reform. *Journal of Public Administration Research and Theory* 11, 165–201.

Musheno M, Maynard-Moody S (2015) 'Playing the Rules': Discretion in Social and Policy Context. In: Hupe P, Hill M, Buffat A (eds) *Understanding Street-Level Bureaucracy*, pp. 169–186. University of Chicago Press, Chicago.

NOP (National Organic Program), 7 C.F.R. § 205 (2016).

Nielsen VL (2006) Are Street-level Bureaucrats Compelled or Enticed to Cope. *Public Administration* 84, 861–889.

Nielsen VL (2007) Differential Treatment and Communicative Interactions: Why the Character of Social Interaction is Important. *Law & Policy* 29, 257–283.

Nielsen VL (2015) Law Enforcement Behavior of Regulatory Inspectors. In: Hupe P, Hill M, Buffat A (eds) *Understanding Street-Level Bureaucracy*, pp. 115–131. University of Chicago Press, Chicago.

Onwuegbuzie AJ, Leech NL (2007) Sampling Designs in Qualitative Research: Making the Sampling Process More Public. *The Qualitative Report* 12, 238–254.

Organic Foods Production Act of 1990, 21 U.S.C. 2101 (1990).

Pautz MC (2009) Trust Between Regulators and the Regulated: A Case Study of Environmental Inspectors and Facility Personnel in Virginia. *Politics & Policy* 37, 1047–1072.

Pautz MC (2010) Front-line Regulators and Their Approach to Environmental Regulation in Southwest Ohio. *Review of Policy Research* 27, 761–780.

Pautz MC, Rinfret SR (2013) *The Lilliputians of Environmental Regulation: The Perspective of State Regulators*. Routledge, New York.

Podsakoff PM, MacKenzie SB, Lee JY, Podsakoff NP (2003) Common Method Biases in Behavioral Research: A Critical Review of the Literature and Recommended Remedies. *Journal of Applied Psychology* 88, 879–903.

Sandfort JR (2000) Moving Beyond Discretion and Outcomes: Examining Public Management from the Front Lines of the Welfare System. *Journal of Public Administration Research and Theory* 10, 729–756.

Schuler RS (1975) Role Perceptions, Satisfaction, and Performance: A Partial Reconciliation. *Journal of Applied Psychology* 60, 683–687.

Selden SC, Brewer GA, Brudney JL (1999) Reconciling Competing Values in Public Administration: Understanding the Administrative Role Concept. *Administration & Society* 31, 171–204.

Talesh S (2015) Rule-intermediaries in Action: How State and Business Stakeholders Influence the Meaning of Consumer Rights in Regulatory Governance Arrangements. *Law & Policy* 37, 1–31.

Teddlie C, Tashakkori A (2009) *Foundations of Mixed Methods Research: Inegrating Quantitative and Qualitative Approaches in the Social and Behavioral Science*. SAGE, Thousand Oaks, CA.

Tyler TR (1990) *Why People Obey the Law*. Yale University Press, New Haven, CT.

Van de Ven AH (2007) *Engaged Scholarship: A Guide for Organizational and Social Research*. OUP, New York, NY.

van der Heijden J (2008) *Competitive Enforcement: Comparative Analysis of Australian Building Regulatory Enforcement Regimes*. IOS Press, Amsterdam, The Netherlands.

Winter SC, May PJ (2001) Motivation for Compliance with Environmental Regulations. *Journal of Policy Analysis and Management* 20, 675–698.

© 2017 John Wiley & Sons Australia, Ltd

Appendix I: Survey respondent sample characteristics

Table A1 Survey respondents by employment status and sectoral affiliations

	Government	Both government and private	Private	Total
Contract	2	5	56	63
Full/part-time employee	9	1	17	27
Total	**11**	**6**	**73**	**90**

Table A2 Survey respondents by years as organic inspectors

1 year	3
2 years	7
3 years	4
4 years	6
5 years	5
6 years	4
7 years	5
8 years	9
9 years	3
10 years	2
> 10 years	44
Total	**92**

Table A3 Survey respondents by gender

Male	41
Female	51
Total	**92**

Table A4 Survey respondents by age

Under 25 years old	1
26–35 years old	10
36–45 years old	15
46–55 years old	22
56–55 years old	32
> 65 years old	12
Total	**92**

Appendix II: Inspector interview questions

How did you come to be an organic inspector?

We identified you as an inspector, but can you describe other roles you have held in the organics industry?

How long have you been conducting organic inspections?

We are aware that inspection of an operation requires knowledge of the products or processes in question. Please describe to me your areas of expertise, and the kinds of inspections you regularly conduct.

Are you a contract, part-time, or full-time employee?

 © 2017 John Wiley & Sons Australia, Ltd

What certifier(s) do you currently work for?

Please describe to me any professional training required by the certifier(s) you work for, and indicate whether the training is provided by the certifier(s) or whether you must obtain it on your own.

Please describe the inspection protocol and required procedures established by your certifier.

How often, if at all, are you in contact with the certifier during an inspection?

How often, if at all, does your certifier have another person accompany you during an inspection?

What materials or information does your certifier provide you for use during an organic inspection?

Reflecting on the various materials and information we have discussed, please describe to me how these impact your regular procedures and practices during an inspection.

How often, if at all, do you interact with other organic inspectors?

From our perspective, an inspector can be considered to be performing three functions: they are a regulatory monitor of NOP regulations, an employee and representative responsible to a certifier, and they are providing a service to the producer or operation under inspection. Please comment on whether you perceive yourself to be operating in these, or different, roles.

How much, and in what manners, do you perceive yourself to have discretion within your inspector functions?

When you come across what you would consider to be minor or trivial non-compliances, how often do you report these to the certifier?

What factors most influence your decision to report or not report minor non-compliances?

Reflecting on the organic operations that you certify, please describe any systematic differences you have observed in levels of compliance with NOP regulations. For example, are there certain characteristics such as size or scope that seem to be associated with compliance levels?

How does an operation's level of compliance impact your inspection practices?

Is there any other information regarding organic inspection that would you like to share with me that we have not already discussed?

Notes

1 Operations that earn less than US$5,000 a year from organic product sales are exempt from this requirement.
2 Another 47 certifiers are based internationally.
3 See interview instrument in Appendix for more details.
4 While not a central focus of this paper, preliminary analyses of survey findings explored differences in inspectors' role perceptions and discretionary attitudes according to their employment status (employee vs. contract inspectors) and organizational affiliations (governmental vs. nongovernmental). The analyses suggested no apparent relationships between inspectors' employment characteristics and their role perceptions and attitudes. No statistically significant differences in role perceptions were found between employee and contract inspectors, or between inspectors working for governmental certifiers and their nongovernmental counterparts. Likewise, no statistically significant differences in discretionary attitudes were found between employee and contract inspectors, or between inspectors working for governmental certifiers and their nongovernmental counterparts. Although these findings do not eliminate the possibility of bias in interview findings as a result of overrepresentation among governmental employee inspectors, they suggest any bias will be limited.
5 The employee/representative and service provider role perceptions are moderately correlated ($r = 0.34$, $P < 0.05$). No correlation was found between the regulatory monitor and employee/representative roles ($r = -0.05$) or the regulatory monitor and service provider roles ($r = 0.05$).
6 Interview ID1.
7 Interview ID4.
8 Interview ID10.
9 Interview ID4.
10 Interview ID1.
11 Interview ID4.
12 Interview ID2.
13 Interview ID10.
14 Interview ID1.
15 Interview ID8.
16 Interview ID2.
17 Interviews ID4, ID10.
18 Interview ID5.
19 Interview ID4.

© 2017 John Wiley & Sons Australia, Ltd